Biology / ESS Field Day Aims 2026

1. Why study the environment?
2. How do scientists collect data in the environment?
3. What is an environmental Impact assessment?
4. How do we measure biodiversity?
5. What are biological indicators?
6. What are human impacts on the environment?

Biology / ESS Field Day Aims 2023

Surveying: Quadrating, transect sampling, identification of species using classification keys, biodiversity

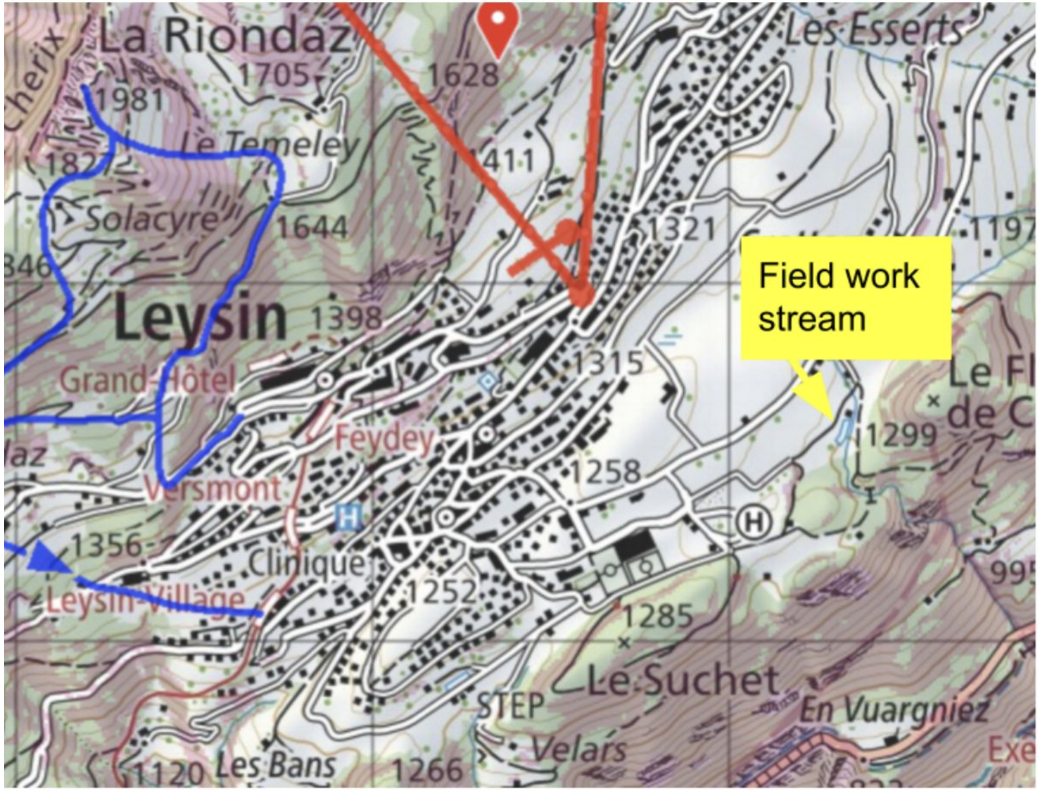
ESS: 2.5 and 3 Quadrating and field work techniques

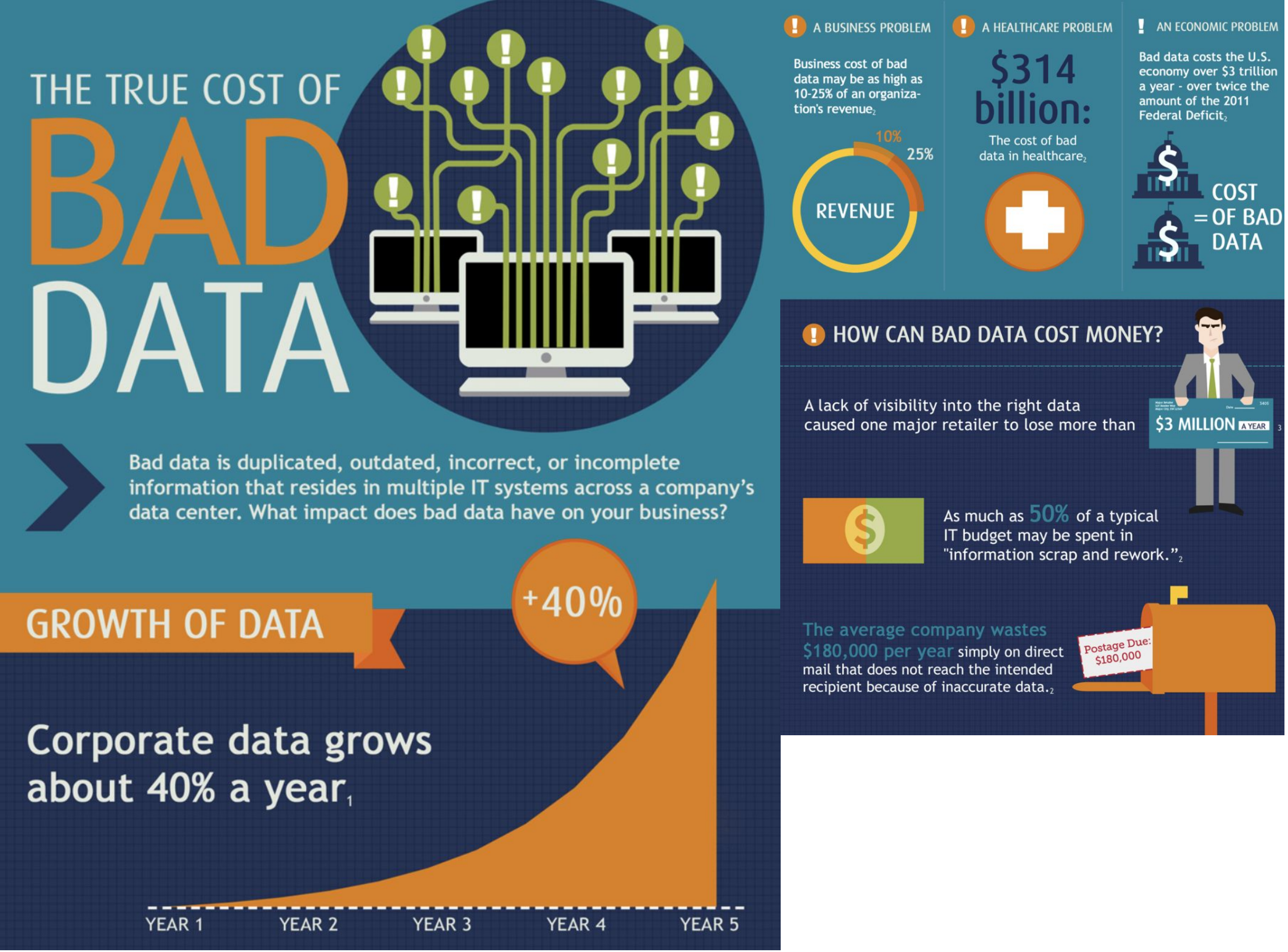
- Biodiversity
- Bioindicators
- Classification keys
- Transects and quadrating
- Data collection
- Simpsons diversity index and biotic index

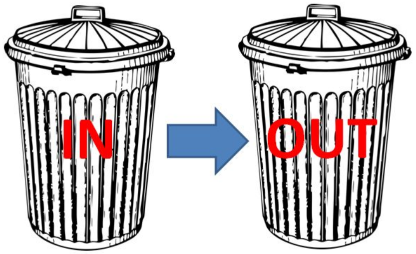
Local Environmental Stream Survey Leysin

Your work will contribute to open source citizen science data

Every year students collect data from the local environment. This data contributes to global knowledge about climate change and is available for use by the scientific community using ARCGIS survey 123 which allows anyone to access the data




THE TRUE COST OF
BAD
DATA
Bad data is duplicated, outdated, incorrect, or incomplete information that resides in multiple IT systems across a company's data center. What impact does bad data have on your business?
GROWTH OF DATA
+40%
Corporate data grows about 40% a year[1]
YEAR 1
YEAR 2
YEAR 3
YEAR 4
YEAR 5
A BUSINESS PROBLEM
Business cost of bad data may be as high as 10-25% of an organization's revenue[2]
10%
25%
REVENUE
A HEALTHCARE PROBLEM
$314 billion:
The cost of bad data in healthcare[2]
AN ECONOMIC PROBLEM
Bad data costs the U.S. economy over $3 trillion a year - over twice the amount of the 2011 Federal Deficit[2]
COST = OF BAD DATA
HOW CAN BAD DATA COST MONEY?
A lack of visibility into the right data caused one major retailer to lose more than
$3 MILLION A YEAR[3]
As much as 50% of a typical IT budget may be spent in "information scrap and rework."[2]
The average company wastes $180,000 per year simply on direct mail that does not reach the intended recipient because of inaccurate data.[2]
Postage Due: $180,000


IN
OUT

Ecology Field Day Form May 2025 All changes saved in Drive

Questions Responses Settings

Section 1 of 3

Ecology Field Day Data Collection Form May 2025

Data collection is a critical part of a scientists' job and doing it well can mean the difference between understanding a problem or being clueless to one.

One of the main objectives of today is for you to reflect on how easy or hard it is to ensure you have "good data".

Recording data reliably in the field is one of the biggest challenges, particularly in natural environments where there may be difficult access, living species and potentially harsh conditions. This form is our tool to make sure we have a store of sufficient and accurate data. EVERYONE'S CONTRIBUTIONS ARE VITAL FOR THE SUCCESS OF THE DAY IF WE WANT "GOOD DATA".

After section 1 Continue to next section

Section 2 of 3

Stream Survey 10am-midday

Description (optional)

What is the number for your stream site

1. 1

ArcGIS Survey123

Included with ArcGIS user types

Smart forms, better decisions

Ecology Field Day Data collection sheet (Google form is available as well)

Stream Survey 10am-midday

Evaluation Question: What is a limitation of collecting field data if you needed to explain exactly where you collected your sample from? How would you improve this?

Click on this Google Map called "LAS Science Map" and add a marker to your site using the GPS function on your phone.
Write in your GPS coordinates below

Steam site data

1. Water temperature (°C)
2. Stream depth (cm)
3. Stream width (cm)
4. What is the velocity using a cork in m/s?
Measure a 1 m stretch and record the time it takes the cork to travel that distance
5. pH
6. Nitrate Level (ppm)
7. Oxygen concentration (ppm) You will need to ask Jiyou to take a reading for you

Species Kick Sample data

	How many?
Mayfly nymph	
Stonefly nymph	
Caddisfly uncased larvae	
Cased caddisfly larvae	
Beetle larvae	
Water penny	
Flat worm	
Damsel fly	
Fly/blackfly larvae	
Shrimp / scud	
Snail	
Leech	
Segmented worm	

Is there any other qualitative data at your site worth recording that could relate to water quality?

Stream Survey Leysin 2023

9/21/22 - 4/29/25 Filter Navigation Print current view

Navigation

Filter questions

- What is your site number for the stream?
- Image of your site
- Is it a pool or a riffle?
- Temperature of water deg.C
- What is the velocity of the water at this site using the cork?
- Depth cm
- Width of stream cm
- pH value
- Nitrate levels
- Oxygen levels ppm
- mayfly nymph
- stonefly nymph

Stats	Value
Min.	1
Max.	10
Avg.	3.642857142857143
Sum.	51

Answered: 14 Skipped:

Image of your site * Gallery

Images: 14

Summary of the day

- 8-10 groups of 2 (or 3)
- You will be measuring stream and meadow
- You will collect abiotic and biotic stream data on bioindicator species for biotic index calculations
- You will collect frequency of organisms we can use for a simpson's diversity index

Plan for the day	
9.15	Meet in SS14
9.30	Bus to lower sporting
10.00 12.00	Stream Ecology Observation skills: Tar spot fungus search + discuss- Chris W/Rachel. Identifying species- iSeek+Birdnet / dichotomous key Questions and reflection time by the fire pit.
12.00-13.00	BBQ lunch yurt
13.00-15.00	Meadow and impact of farming
15.00	Bus return to school with equipment meet in lab to disseminate information
15.25	finish at end of school day

Things that get your points

-Winning the challenges we set

-Having completed data sets that are legible and well organised

-

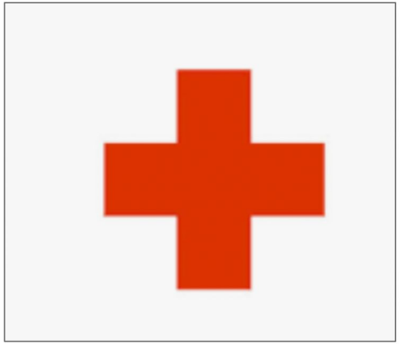
What to bring

***backpack or borrow one** 1 per group of 3
***long socks** (protection from ticks)
***walking shoes/trainers** (walking in the forest requires sturdy footwear, it is steep and rocky)
***long trousers** (there might be ticks)
***waterproof jacket** (it might rain or for protection against wind)
***extra layers of clothing/hat / gloves** (if it is cold on the day)
***water bottle** (stay hydrated)
***iPhone** (must be fully charged)

Health and Safety

On the way to the plot:
Traffic - pay attention on the road, walk in two's
Cows - walk as a group, don't go too close
Walking ground - pay attention to the ground under your feet, take care to place your feet

In the plot:
Sticks in eyes - wear eye protection in the plot
Steep ground - walk carefully and slowly around
Stings - we have cream
Cuts - pay attention
Sun exposure - apply cream, wear a hat and glasses
Dehydration - bring and drink water
Ticks - long trousers, socks, repellant



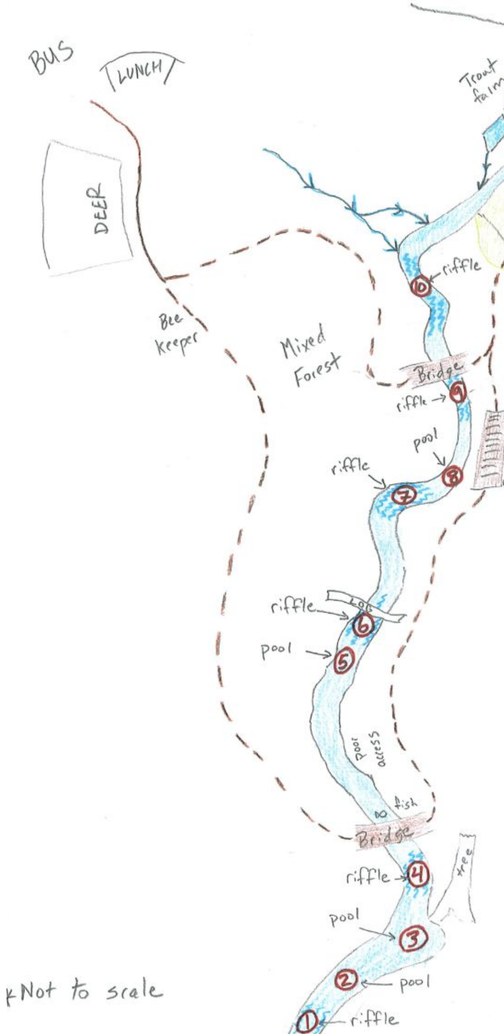
We carry a first aid kit with: bandages, plasters, insect repellant, suncream

Stream survey: Abiotic & Biotic

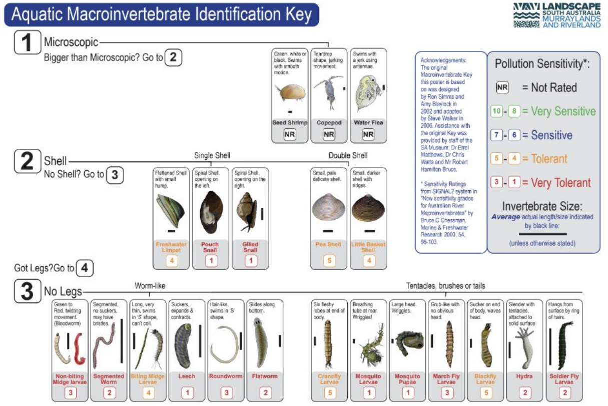
When we want a measure of the **impact of human activity** such as fish farming or road works on the indicator species in a stream we monitor numbers and then over time we can see what is happening in the stream.

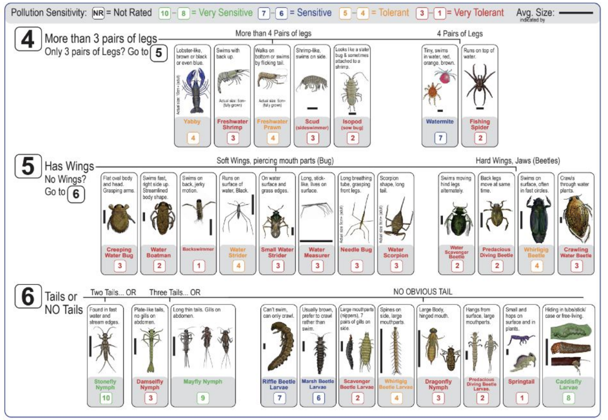
We will measure abiotic factors such as:

- pH
- temperature
- Velocity
- Stream width and depth
- Nitrate levels



Stream survey: Biotic indicators and a biotic factors





Physico-chemical Indicators (~15 mins)

What position on the stream are you? (1-10)

Type of water flow	*pool/riffle			
	1	2	3	mean
Velocity				
Width of river				
Depth of river				
pH				
Temperature				
Nitrates				

We need to REPEAT the method for reliability

We need to STANDARDISE the method for accuracy

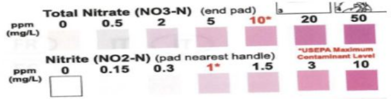
And to make the data comparable **between sites** comparable between **different years**

Read how to do this below …

How to measure pH
pH - place the probe the middle of the water column
Check this with the litmus paper indicator

How to measure depth and width
Hold the meter ruler in the water at three location on your site **vertically touching the stream bed.** Measure where the water line is **most of the time over 30 seconds**
Measure three widths of the stream at the edge of the water where it meets the bank horizontally

Nitrates
Place stick in **middle** of water column for **2 seconds** *Do not move it about*
Wait **1 minute** check colour against chart



How to measure velocity
1. Place a meter ruler in the water
2. Hold a phone above the ruler and video
3. Release creamer and film the movement holding the phone still
4. Calculate the time taken for the creamer to move 1m - repeat as necessary
5. velocity = distance/time

How to measure temperature
Hold the thermometer in the **middle** of the water column for **30s**
Read the temperature whilst **still in the water**

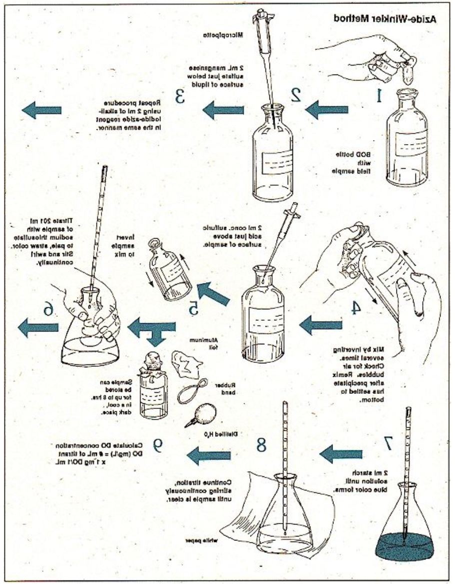
If analogue - read at at eye level (get down to the thermometer)

Soil bags

Water samples

To test for levels of contaminants- nitrates, phosphates, Metals ions

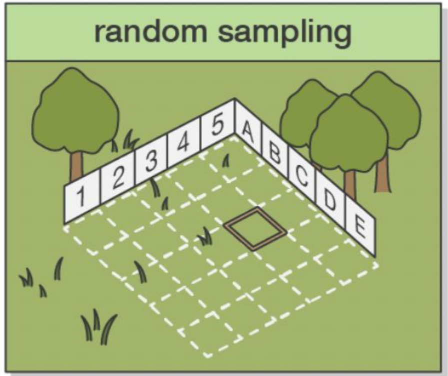
To test for dissolved oxygen levels using the Winkler Method

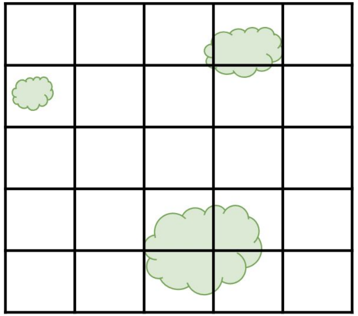


Meadow: Marsh V Drained agriculture

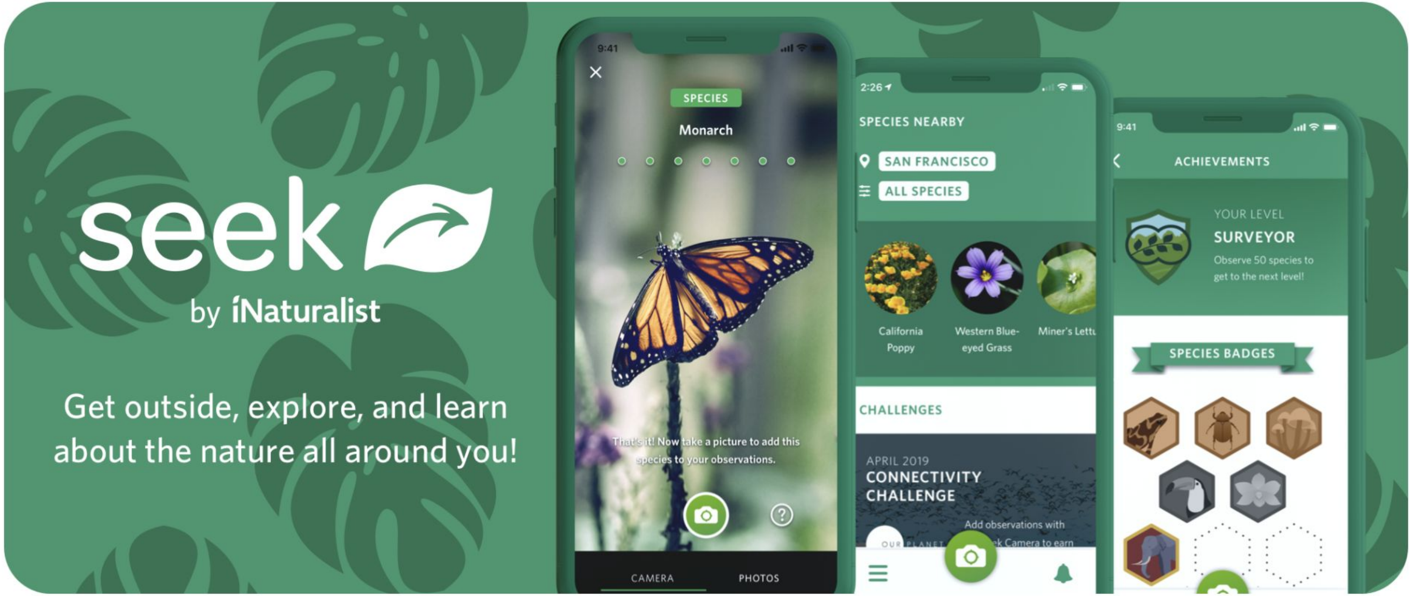
We will carry out a random sample of the plant life and insect life in two sites

- Agree on the species labels A, B, C, D etc on a card and take a picture
- Lay out two 30m tape measures
- Take a 1x1m quadrat
- Find a random coordinate
- Lay the top left corner of the quadrat at that coordinate
- Count the number of individuals for each species in the quadrat as percentage cover





Seek by iNaturalist



GET IT ON Google Play

Download on the App Store

Reflection: Photovoice instructions

What is Photovoice? a playful means of supporting your reflection about your experience of the day. How you feel at the time and how you describe and feel after the event.

What to do? During the day, please take pictures **with your mobile** – as many as you want – of significant events/moments of the day

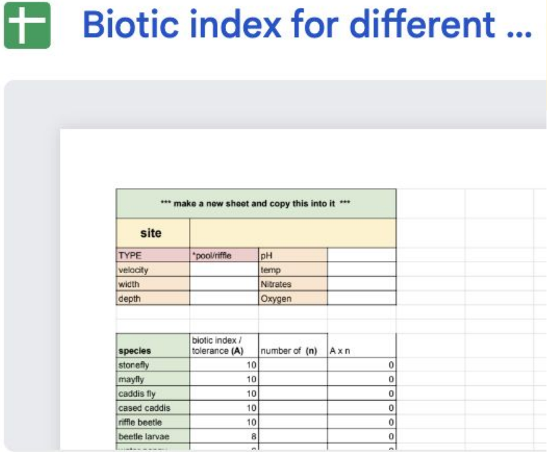
NB: Don’t worry about the artistic quality of your pictures. What matters is what the content of the photos **means to you**.

Now complete the photovoice

Back in the classroom stream

- Calculate the biotic index for pool V riffle by inputting the data

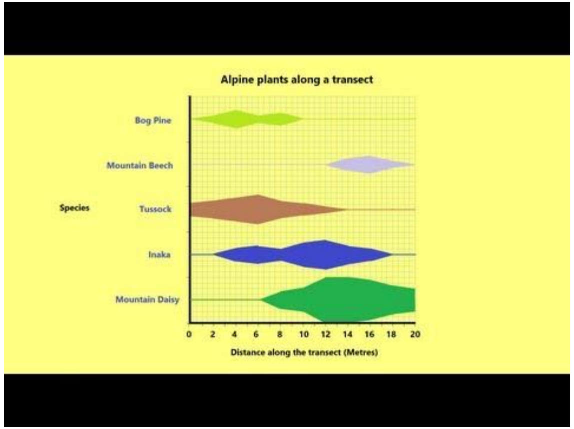
- Create kite diagrams to show one abiotic factor against species numbers for at least 3 indicator species


Biotic index for different ...

*** make a new sheet and copy this into it ***

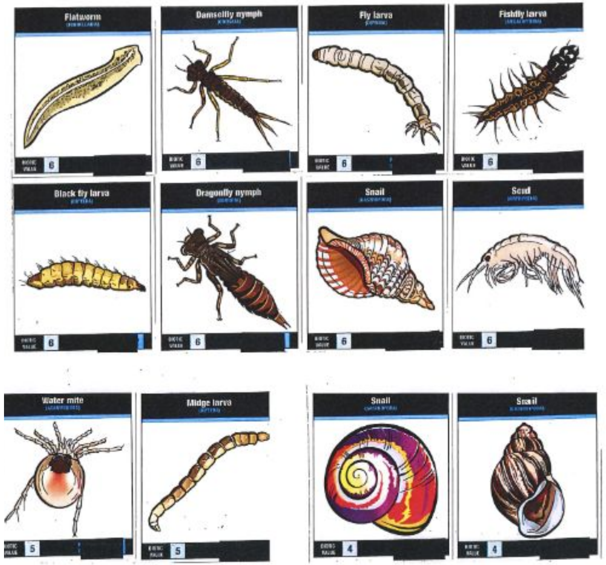
site			
TYPE	*pool/riffle	pH	
velocity		temp	
width		Nitrates	
depth		Oxygen	

species	biotic index / tolerance (A)	number of (n)	A x n
stonefly	10		0
mayfly	10		0
caddis fly	10		0
cased caddis	10		0
riffle beetle	10		0
beetle larvae	8		0



Stream survey Biotic indicators - using the data collected work out the biotic index for pool and riffle and relate to oxygen levels

	0	1-5	6-20	21-50	50-99	over 100
mayfly nymph*						
stonefly nymph*						
caddisfly larvae uncased*						
caddisfly cased*						
riffle beetle*						
water penny*						
flat worm*						
damselfly*						
fly/black fly larvae*						
snail*						
shrimp / scud*						
leech*						
segmented worm*						



$$\text{Biotic Index} = \frac{\sum (n_i \times a_i)}{N}$$

KEY →

- N = Total number of individuals collected
- n_i = Number of individuals of a species
- a_i = Tolerance rating of a species

Indicator Species

Pollution Tolerance and Environmental Pollution Levels

Indicator species	Category	Rating
Stonefly nymph 	**Class I** Pollution sensitive	9
Freshwater shrimp 	**Class II** Moderately tolerant	5
Tubifex worm 	**Class III** Pollution tolerant	1

Biotic Index	Water Quality	Pollution Level
> 10	**Good** Clean stream	Organic pollution unlikely
3 – 9	**Average** Some pollution	Moderate organic pollution likely
0 – 2	**Poor** Gross pollution	Substantial organic pollution

Back in the classroom meadow

Calculate simpsons diversity index

- Create a graph of one abiotic factor against simpsons